## Quotation No : Q24000495

Date : 24-Sep-2024

Simar Enterprises
Ground Floor, 76, Transport Center, Punjabi Bagh
NEW DELHI
DELHI

## Kind Attention : Mr./Ms. Karan Wadhwa

Dear Sir,

We understand that in industry, reliability is not just a requirement,it is the foundation of success.
That is why Absstem Industrial PSA Generator is more than just a product; it is a commitment to excellence, efficiency, and uninterrupted performance.

Every component is designed with precision and innovation, ensuring that each molecule of oxygen and nitrogen is delivered with the utmost purity.

With fully automated operation, our system seamlessly integrates into your processes, reducing manual intervention and maximizing productivity.

Moreover, our industry-leading air ratio ensures that you achieve the highest efficiency, optimizing energy consumption while maintaining peak performance.
And because we believe in standing by our craftsmanship, we offer an unmatched-10 year warranty on sieves, giving you the peace of mind that your investment is protected for the long run.

We have attached the detailed quotation for your review.

Should you have any questions or require further customization to meet your specific needs, please do not hesitate to reach out.

We are here to provide not just a product but a solution that breathes life into your operations.

We look forward to the opportunity to serve you and build a lasting partnership.

Thanks and Regards,

## For Absstem Industrial Solutions Pvt. Ltd.

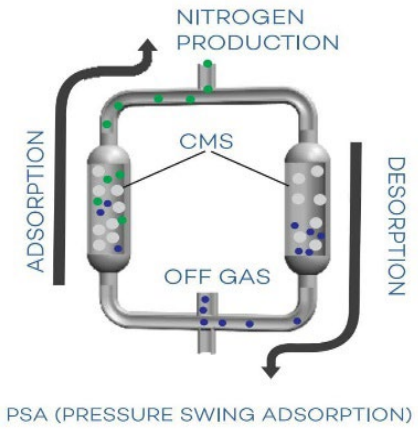
Arifa Zaidi
Assistant Manager Business Development
9218108184
www.absstem.com

nGEN POWERED BY ABSSTEM INDUSTRIAL

# Industrial PSA Nitrogen Generator




## PSA Technology




PSA (PRESSURE SWING ADSORPTION)

Industrial Nitrogen Generator mainly consists of two adsorbers filled with carbon molecular sieves. Pressurized air is passed through one of these adsorbers, carbon having the property to attract oxygen, adsorbs oxygen from air and Nitrogen is allowed to pass through and move into the storage tank as the final product. As the adsorber approaches saturation, the air is allowed to pass through the other adsorber while this adsorber is being regenerated to ensure a continuous supply of Nitrogen. This changeover takes place through a specially designed microcontroller to ensure a hassle-free operation. The quality of Nitrogen produced is monitored by a Nitrogen analyser.

## Key Features

- Shipped ready for installation
- 5 minutes to run and never have to stop it!
- Easy installation
- Low operating expenses compared to gas cylinders/liquid Nitrogen
- Maintenance costs are very less
- Embedded Safety Interlocks
- Space-saving design
- Eliminates all hazards which are there with gas cylinder or liquid Nitrogen
- Longevity: at least 10 years

**Absstem Industrial Solutions Pvt. Ltd.**
Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122051
Contact - 1800 3010 3394 Email - indsales@absstem.com
CIN: U29256HR2022PTC103387

## Description

| | | |
|---|---|---|
| a. | Product Name | Nitrogen Gas Generator |
| b. | Model Range | **NGEN** |
| | Technology | PSA (Pressure Swing Adsorption) |

## Specifications

| | | |
|---|---|---|
| **a.** | Nitrogen Flow Rate | As per Performance Sheet |
| **b.** | Nitrogen Purity | As per Performance Sheet |
| **c.** | Nitrogen Delivery Pressure | 5.5 Barg (Higher Pressure Options Available) |
| **d.** | Ambient conditions | |
| | Temperature | 0 to 50 Deg. Celsius |
| | Relative Humidity | up to 80 % (Max.) |
| | Altitude | up to 1000 m above Sea level |

## Scheme Offered – (PSA Only)

This scheme will comprise of

- Air package with inlet air filtration system
  (Air compressor, Air dryer, Water Separator, Oil filters)
- Industrial Nitrogen generator
- Storage Tank
- Nitrogen Analyzer
- Alarm panel

## Unique Features of Absstem

a. All the fabricated items are cleaned and degreased following our strict quality policy.
b. All the Instruments are specially manufactured to make them suitable for Nitrogen gas.
c. Guaranteed 10+ years of healthy life.
d. We provide the purest **Nitrogen** in Industry.
e. Data logging and Graphical Reports for all parameters available.
f. Test certificates of all instruments are available on demand.

## **Onsite Gas Generation** vs Liquid Tank/Cylinder Alternative

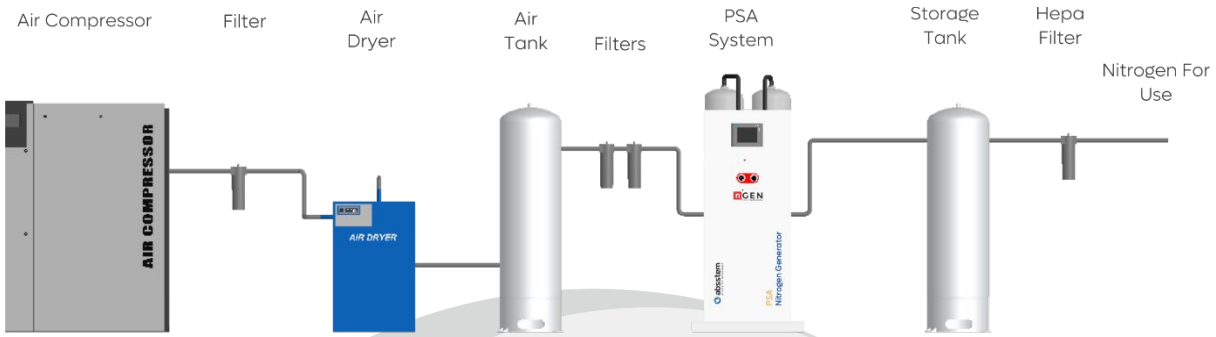
| |
|---|
| *Non-stop availability: 24 hours a day, 7 days a week* |
| *Easy integration within existing installations* |
| *Your own independent supply of industrial gas* |
| *Significant economies for scale and lower operational costs: no rental charges, logistical expenses and bulk user evaporation losses* |
| *No safety hazards when handling high-pressure cylinders* |



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## Equipment Description

The Industrial Nitrogen Generator consists of




| | |
|---|---|
| Air Compressor | An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Nitrogen** from the air. |
| Refrigeration dryer | Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air. |
| Air Tank | Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system. |
| Multistage Filtration system | Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels. |
| PSA system | Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Nitrogen** from Air. |
| Storage Tank | Industrial Grade **Nitrogen** produced from NGEN is stored in this tank to supply a continuous flow of **Nitrogen**. |
| Nitrogen Analyzer | Nitrogen produced is monitored by a **Nitrogen** analyser. In case, O2 purity goes below the specified level, it gives an alarm and switches it to vent mode. |

## HD Touch Screen Display

(User-friendly touchscreen interface for easy operation and maintenance)

Absstem industrial Nitrogen generator complies with all industry standards and regulation

- Alarm management system
- Plug and play
- Fully automatic
- Process data display
- Consumables running hours display
- Data logging via USB drive
- Graphical representation




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## Additional upgrades

- Digital flow meter with totaliser
- High-pressure Nitrogen tank
- Cylinder manifold and alarm management system
- Remote monitoring

## Plant Operation

- Unit operates 24-hrs a day, 7-days a week reliably, automatically & safely

## Automation

- The Plant is fully automatic and needs only electrical power for running the Air compressor
- The machine automatically enters into standby mode when the level in the Storage Tank reaches its set limit. This helps to increase the life of the Air compressor.

## International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Nitrogen generators.

## Itemised Supply Description

| No. | Item description | Make | Qty |
|---|---|---|---|
| a | Air Compressor with Filters | ABSSTEM APPROVED | Quoted Separately |
| b | Refrigeration Dryer | ABSSTEM APPROVED | Quoted Separately |
| c | Air Tank | ABSSTEM | Quoted Separately |
| d | Set of Filters | ABSSTEM | 1 no. |
| e | PSA unit (NGEN) | ABSSTEM | 1 no. |
| f | Electrical Power Panel with PLC and all cabling | ABSSTEM & MITSUBISHI | 1 no. |
| f | Digital Nitrogen Analyzer | ABSSTEM | 1 no. |
| g | Storage Tank | ABSSTEM | Quoted Separately |
| h | Outlet product gas Filter | ABSSTEM | 1 no. |



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## Electrical Protection

The client should make proper arrangements for earthing/grounding systems for the Industrial **Nitrogen** Generator at their facility.

## Enclosure Panel

- Ergonomically designed metal enclosure with powder-coated finish for longer life.
- High-strength IP-rated panel fittings from Germany for a robust enclosure.
- Fan filters from **Rittal** are IP-55 rated making the electrical enclosure's internal environment dust-free.
- Easy connectivity to any place of installation with req. port connections.
- High-strength base frame designed for higher load and easy transportation.
- Vessels are designed per ASME section VIII DIV I. Vessels are epoxy paint coated which is considered good for longevity and corrosion resistance.
- All electrical items are CE certified with operating voltage at 24 VDC

## Annual Maintenance Contract (AMC)

Our in-house support Absstem shield is 24*7 open and we assure a timely service maintenance schedule. Per the schedule mentioned in the operational manual, we suggest the operator change the filters by following a simple protocol.

It is very easy to change or replace as our commissioning engineer plus our commissioning engineer and tech support office will always guide you about the activity.

## Best ROI Guaranteed

***ROI within 6 to 18 months when compared to conventional methods***

Generator performance depends on the component selection. Our processes make this job easier as all equipment is selected based on more than 3 decades of research. NGEN is a fully automatic (operator-free operation) that ensures peace of mind.

# We deliver what we commit!!

Thank You



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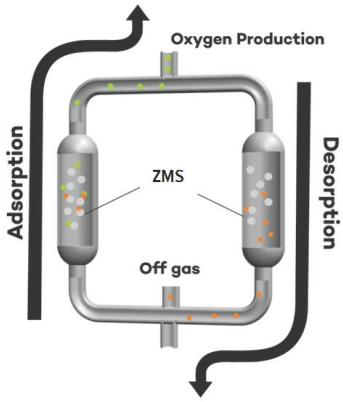
POWERED BY ABSSTEM INDUSTRIAL

# Industrial

# PSA Oxygen Generator




## PSA Technology




PSA Process explaining oxygen production

Industrial Oxygen Generator mainly consists of two adsorbers filled with zeolite molecular sieves. Pressurized air is passed through one of these adsorbers, zeolite having a property to attract nitrogen, adsorbs Nitrogen from air and oxygen is allowed to pass through and move into the storage tank as the final product. As the adsorber approaches saturation, the air is allowed to pass through the other adsorber while this adsorber is being regenerated to ensure a continuous supply of Oxygen. This changeover takes place through a specially designed microcontroller to ensure a hassle-free operation. The quality of oxygen produced is monitored by an Oxygen analyser.

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*CIN -U29256HR2022PTC103387*

## Key Features

- Shipped ready for installation
- 5 minutes Startup time with uninterrupted stoppages.
- Quick & Easy installation
- Low operating costs compared to gas cylinders/liquid oxygen
- Reduced Maintenance costs
- Embedded Safety Interlocks to cater any abnormality.
- Space-saving compact design.
- Eliminates all safety hazards which are there with gas cylinder or liquid oxygen.
- 10 years warranty on molecular sieves.

## Description

| | |
|---|---|
| a. Product | Oxygen Gas Generator |
| b. Model | **IDOS** |
| c. Technology | PSA (Pressure Swing Adsorption) |

## Specifications

| | | |
|---|---|---|
| **a.** | Oxygen Flow Rate | As per Performance Sheet |
| **b.** | Oxygen Purity | As per Performance Sheet |
| **c.** | Oxygen Delivery Pressure | 5.5 Barg (Higher Pressure Options Available) |
| **d.** | Ambient conditions | |
| | • Temperature | 0 to 50 Deg. Celsius |
| | • Relative Humidity | up to 80 % (Max.) |
| | • Altitude | up to 1000 m above Sea level |

## Scheme Offered

This scheme will comprise of

- Air package
  (Air compressor, Air dryer, Air Receiver, Water Separator)
- Industrial Oxygen generator with oil removal filtration system
- Storage Tank
- Oxygen Analyzer
- Alarm panel

## Unique Features of Absstem

a. All fabricated items are cleaned and degreased as per GEP standards.
b. All Instruments used are compatible with oxygen gas.
c. Guaranteed 10+ years of trouble free and reliable plant operation.
d. We provide the purest **Oxygen** in Industry.
e. Data logging and Graphical Reports for all parameters available.
f. Test certificates of all instruments are available on demand.

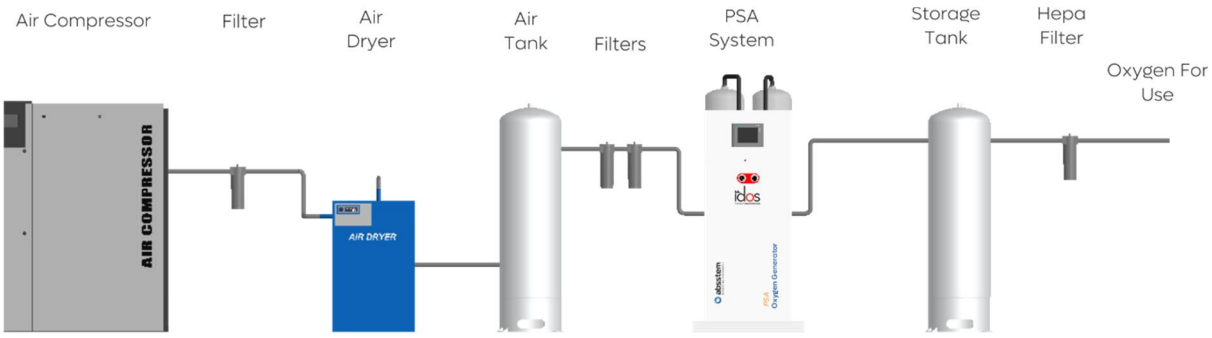
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*CIN -U29256HR2022PTC103387*

# **Onsite Gas Generation** vs Liquid Tank/Cylinder Alternative

| |
|---|
| *Non-stop availability: 24 x 7 across 365 days in an year* |
| *Easy integration within existing installations* |
| *Your own independent supply source of industrial gas* |
| *Significant economies for scale and lower operational costs: no rental charges, logistical expenses and bulk user evaporation losses* |
| *No safety hazards related to handling high-pressure cylinders* |

## Equipment Description

The Industrial Oxygen Generator consists of




| | |
|---|---|
| Air Compressor | An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **oxygen** from the air. |
| Refrigerated Air Dryer | Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air. |
| Air Receiver | Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system. |
| Multistage Filtration system | Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels. |
| PSA system | Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Oxygen** from Air. |
| Storage Tank | Industrial Grade **Oxygen** produced from IDOS is stored in this tank to supply a continuous flow of **Oxygen**. |
| Online Oxygen Analyzer | Online Oxygen analyzer for continuous monitoring of Oxygen purity level. |

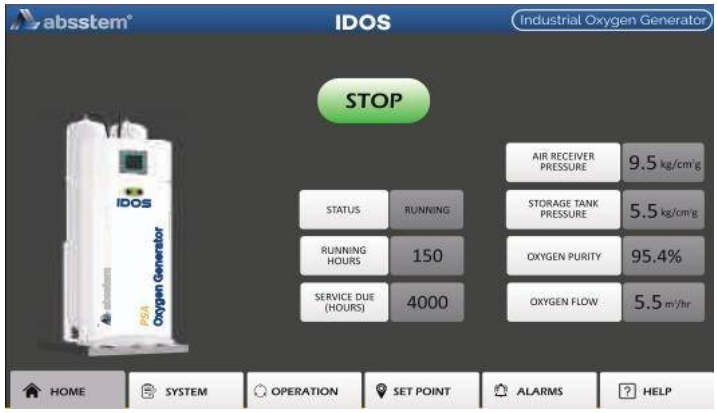
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*CIN -U29256HR2022PTC103387*

## HD Touch Screen Display

(User-friendly touch screen interface for easy operation and maintenance)

Absstem industrial oxygen generator complies with all industry standards and regulation

- Alarm management system
- Plug and play
- Fully automatic
- Process data display
- Consumables running hours display
- Data logging via USB drive
- Graphical representation




## Additional upgrades

- Digital flow meter with totaliser
- High-pressure oxygen tank
- Oxygen booster for high pressure requirements
- Cylinder manifold and alarm management system
- Remote monitoring

## Plant Operation

- Unit operates 24-hrs a day, 7-days a week throughout the year reliably, automatically & safely

## Automation

- The Plant is fully automatic and needs only electrical power for running the Air compressor
- The machine automatically enters into standby mode when the level in the Storage Tank reaches a preset limit. This helps in optimised Air Compressor operation and reduced power consumption.

## International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial oxygen generators.

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*CIN -U29256HR2022PTC103387*

## Itemised Supply description

| No. | Item description | Make | Qty |
|---|---|---|---|
| a | Air Compressor | Kaeser/IR/As per Demand | Client scope/ Quoted Separately |
| b | Refrigeration Dryer with pre filter | Kaeser/IR/As per Demand | Client scope/ Part of Compressor Package. |
| c | Air Tank | ABSSTEM | Client scope/ Part of Compressor Package. |
| d | Set of Filters | ABSSTEM | 1 no. |
| e | PSA unit (IDOS) | ABSSTEM | 1 no. |
| f | Electrical Power Panel with PLC and cabling | ABSSTEM & MITSUBISHI | 1 no. |
| f | Digital Oxygen Analyzer | ABSSTEM | 1 no. |
| g | Storage Tank | ABSSTEM | Quoted Separately. |

## Electrical Protection

The client should make proper arrangements for earthing/grounding systems for the Industrial **Oxygen** Generator at their facility.

## Enclosure Panel

- Ergonomically designed metal enclosure with powder-coated finish for longer life.
- High-strength IP-rated panel fittings from Germany for a robust enclosure.
- Fan filters from Rittal are IP-55 rated making the electrical enclosure internal environment dust-free.
- Easy connectivity to any place of installation with req. port connections.
- High-strength base frame designed for higher load and easy transportation.
- Vessels are designed per ASME section VIII DIV I. Vessels are epoxy paint coated which is considered good for longevity and corrosion resistance.
- All electrical items are CE certified with operating voltage at 24 VDC

## Maintenance Contract (AMC / CMC)

Our in-house service support packages Absstem Shield & Absstem Shield premium ensures 24*7 on site support for all scheduled and need based service requirements. Per the schedule mentioned in the operational manual, we suggest the user to change the filters by following a simple protocol.

Our technical experts are always available to guide your maintenance team for carrying out minor troubleshooting and basic maintenance activities via remote support program.

## Best ROI Guaranteed

***ROI within 6 to 18 months when compared to conventional methods***

Generator performance depends on the component selection. Our processes make this job easier as all equipment is selected based on more than 3 decades of research. IDOS is a fully automatic (operator-free operation) that ensures peace of mind.

## We deliver what we commit!!

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*Contact-1800 3010 3394, Email- Indsales@absstem.com*
*CIN -U29256HR2022PTC103387*

# Product Description

| Product | Description | Make | Qty |
|---|---|---|---|
| nGen Nitrogen Generator | nGEN 01<br>Nitrogen | | 1 |

| Caption | Specification |
|---|---|
| Gas Generated | Nitrogen |
| Gas Pressure | upto 6.5 Barg( Higher options Available) |
| Flow @ 95 % | 9 Nm3/hr |
| Flow @ 97 % | 8 Nm3/hr |
| Flow @ 98 % | 7.5 Nm3/hr |
| Flow @ 99 % | 7 Nm3/hr |
| Flow @ 99.5 % | 6 Nm3/hr |
| Flow @ 99.9 % | 5 Nm3/hr |
| Flow @ 99.95 % | 4 Nm3/hr |
| Compressed Air Requirement | 12 CFM |
| Module | PSA Plant |
| Air Compressor And Drier | In your Scope / Quoted Separately |

| Product | Description | Make | Qty |
|---|---|---|---|
| nGen Gas Storage | nGEN 01 / 02 / 04 ST<br>Nitrogen | | 1 |

| Caption | Specification |
|---|---|
| Water Volume | 500 Liters |
| Gas Stored | Nitrogen |
| Gas Stored Volume | 2.75 M3 |
| Working Pressure | 6.5 Barg |
| Hydrotest Pressure | 9 Barg |



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**Contact**: 1800 547 0506 | indsales@absstem.com
**CIN**: U29256HR2022PTC103387

| Product | Description | Make | Qty |
|---|---|---|---|
| IDOS Oxygen Generator | IDOS 09 B<br>Oxygen | | 1 |

| Caption | Specification |
|---|---|
| Gas Generated | Oxygen |
| Flow | 9 Nm3/Hr |
| Purity | 95 Percent |
| Compressor | In Your Scope / Quoted Separately |
| Dryer | In Your Scope / Part of Compressor Package |
| Air Vessel | In Your Scope / Part of Compressor Package |
| Dry Air Required | 71 CFM |
| Power Required | 230V 50 Hz Single Phase |
| Module Option | PSA Plant Only |

| Product | Description | Make | Qty |
|---|---|---|---|
| IDOS Gas Storage | IDOS 09 ST<br>Oxygen | | 1 |

| Caption | Specification |
|---|---|
| Water Volume | 2000 Litres |
| Gas Stored | Oxygen |
| Gas Stored Volume | 11 M3 |
| Working Pressure | 5.5 Barg |
| Hydrotest Pressure | 9 Barg |

## Commercial Offer

| S No | Item Code | Item Name | Description | HSN | Qty | Unit Price | GST(%) | Amount |
|---|---|---|---|---|---|---|---|---|
| 1 | nGEN 01 | nGEN 01 | nGen Nitrogen Generator | 840510 | 1 | 6,45,600.00 | 18.00 | 6,45,600.00 |
| 2 | nGEN 01 ST | nGEN 01 / 02 / 04 ST | nGen Gas Storage | 840510 | 1 | 72,000.00 | 18.00 | 72,000.00 |
| 3 | IDOS 09 B | IDOS 09 B | IDOS Oxygen Generator | 842139 | 1 | 17,65,200.00 | 18.00 | 17,65,200.00 |
| 4 | IDOS 09 ST | IDOS 09 ST | IDOS Gas Storage | 842139 | 1 | 1,92,000.00 | 18.00 | 1,92,000.00 |
| | | | | | | | Sub Total | 26,74,800.00 |



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**CIN**: U29256HR2022PTC103387

| | |
|---|---|
| GST(%) @18.00 | 4,81,464.00 |
| Grand Total () | 31,56,264.00 |

## Terms And Conditions

| | | |
|---|---|---|
| 1. | EXCLUSIONS & CLIENTS SCOPE | All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clients site will be in Client Scope. Additional gas piping, cooling water (if required). |
| 2. | SUPERVISION OF ERECTION & COMMISSIONING | There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you. |
| 3. | P&F | 3% Extra. |
| 4. | Freight | Extra at Actuals. |
| 5. | Insurance | In your Scope. |
| 6. | Payment Terms | 50% Advance with your Prestegious Purchase Order and Balance against PI. |
| 7. | Delivery | 6-8 Weeks from the PO and Advance. Rest can be confirmed for Availability. |
| 8. | Warranty | We hereby confirm that, our equipment shall have Suppliers warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier. |
| 9. | GST | As per the Commercial Table. |
| 10. | Remarks | . |

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For Absstem Industrial Solutions Pvt. Ltd.

Arifa Zaidi

Assistant Manager Business Development

9218108184

www.absstem.com